

Supplementary Material for PosIR

For detailed information of the models appearing in this paper, please refer to Table 12.

A PosIR Datasets

A.1 Overview

PosIR comprises 310 datasets (17,329,673 documents and 421,708 queries) spanning 10 languages and 31 domains, forming a large-scale multilingual benchmark for information retrieval. For each dataset, we use the same format as BEIR, i.e. corpus, queries and qrels, which will be all available in the Hugging Face Hub. For each dataset, we use the same format as BEIR, i.e. corpus, queries and qrels, which are all available in the Hugging Face Hub.¹³ Table 4 summarizes language-level statistics, showing well-balanced corpus sizes across languages, with each language contributing approximately 1.7M–1.9M documents. Token length statistics reveal substantial cross-lingual variation that reflects differences in writing systems and morphological complexity. Logographic languages exhibit more compact representations (e.g., Chinese and English), whereas morphologically rich alphabetic languages require longer sequences (e.g., Arabic and Russian). Table 5 reports domain-level aggregates and highlights pronounced variation in domain scale and document characteristics. Domain sizes range from several hundred thousand documents to over three quarters of a million, while token length distributions reflect domain-specific content properties. Technical domains such as Information Security tend to contain longer documents, whereas more formal or symbolic domains, such as Mathematics and Statistics, exhibit comparatively shorter texts. This diversity enables evaluation of IR systems under heterogeneous conditions. For comprehensive coverage, Tables 13–43 provide detailed statistics for all 31 domains, presented individually for each domain. These tables report language-specific document counts, query counts, average token lengths, and token length ratios relative to English. Such fine-grained reporting supports reproducibility and facilitates the analysis of domain–language interaction effects in multilingual IR.

A.2 Diversity Analysis

A.2.1 Query Diversity. To analyze the diversity of query types in PosIR, we adopt a keyword-based heuristic that categorizes queries according to their leading interrogative terms (e.g., *WHAT*), with non-interrogative queries grouped as *CLAIM*. Table 6 summarizes the distribution of query types in the English and Chinese retrieval datasets.¹⁴ Several observations can be made from the results. First, *WHAT* queries constitute the largest proportion of the datasets, followed by *HOW* and *WHY* queries, indicating that the generated queries predominantly seek factual explanations and procedural information. Notably, Chinese queries contain a higher proportion of *WHY* and *WHO*, whereas English queries exhibit a larger share of queries categorized as *OTHERS*, suggesting differences in query formulation across languages. In addition, a small fraction of queries

are categorized as *CLAIM*, representing declarative statements that express information needs without explicit interrogative forms.

A.2.2 Corpus Diversity. Following the approach in MTEB [17], we compute inter-domain similarities separately for English and Chinese corpora, each consisting of 31 domains, in PosIR. Specifically, we adopt the highest-scoring model on the MTEB (eng, v2) STS task, QZhou-Embedding (91.65) [30], to generate embeddings for all documents within each domain. We then average document embeddings at the domain level and compute pairwise cosine similarities between domains. The resulting similarity matrix is visualized as a heatmap in Figure 6.

B Human Evaluation of Generated Queries and Reference Spans

B.1 Evaluation Setup

To validate the quality of the generated queries and reference spans in PosIR, we conducted targeted human evaluation on a representative subset. Given the large scale of PosIR (310 datasets spanning 10 languages and 31 domains), we focused our human evaluation on the English subset to enable a focused and thorough assessment. Specifically, we randomly sampled 500 English query-document-reference span triples across 10 diverse domains (50 samples per domain). Each sample consists of: (1) a query generated by LLMs, (2) the corresponding positive document, and (3) the reference span annotated in the qrels, which we evaluated to verify whether it accurately localizes the answer within the document.

B.2 Evaluation Protocol

One annotator evaluated each sample across three dimensions using a structured evaluation form:

- **Query Quality (1-5 Scale):** This dimension assesses the overall quality of the query for retrieval. Scores range from 5 (a perfect query that is clear, specific, fully relevant to the document) to 1 (query with significant issues in relevance, clarity, or answerability).
- **Span Necessity (Binary 0/1):** This dimension evaluates whether the reference span contains only the necessary content. A score 1 indicates the span captures the core answer without irrelevant context, and a score 0 indicates over-extraction with unnecessary surrounding content.
- **Span Sufficiency (Binary 0/1):** This dimension evaluates whether the reference span contains complete information to answer the query. A score 1 indicates the span is sufficient with the core answer completely contained, and a score 0 indicates under-extraction where some supporting details are missing.

We also evaluated whether the generated spans fabricate information not grounded in the document.

B.3 Evaluation Results

The evaluation results demonstrate the high quality of PosIR’s position-aware annotations. Query quality achieved a mean score of 4.91, with 496 of 500 samples (99.2%) rated ≥ 4 , indicating Good or Excellent quality. For span precision, 490 of 500 spans (98.0%) satisfied both necessity and sufficiency criteria. Among the 10 remaining

¹³<https://huggingface.co/datasets/infgrad/PosIR-Benchmark-v1>

¹⁴The *OTHERS* category mainly includes yes/no questions—typically those starting with auxiliary verbs (e.g., *do/does/did*, *can*, *will*) or be verbs (e.g., *is/are*)—as well as other infrequent or implicit question forms that do not match predefined interrogative patterns.

Table 4: Summary of corpus statistics aggregated by language.

Language	Document Count			Token Length					
	Corpus	Queries	Ratio	Min	Q1	Median	Q3	Max	Mean
Arabic	1,712,847	41,894	9.89%	5	717	1401	2250	4719	1509.5
Chinese	1,870,228	43,931	10.78%	13	473	869	1410	2050	950.8
German	1,718,556	41,983	9.92%	3	655	1278	2067	4296	1387.3
English	1,719,701	42,015	9.92%	14	416	842	1392	2051	922.4
French	1,718,832	42,001	9.92%	2	645	1269	2054	4352	1375.1
Italian	1,718,831	42,005	9.92%	3	660	1295	2090	4298	1401.3
Korean	1,715,299	41,913	9.90%	5	704	1403	2258	4685	1509.5
Portuguese	1,718,848	42,002	9.92%	3	607	1195	1941	4020	1298.1
Russian	1,718,056	41,969	9.91%	2	721	1424	2300	4828	1539.2
Spanish	1,718,475	41,995	9.92%	4	604	1187	1925	3956	1287.9
Total/Avg	17,329,673	421,708	100.00%	5	620	1216	1968	3925	1318.1

cases, 3 exhibited over-extraction and 7 exhibited under-extraction. Notably, no hallucinations were observed. These results confirm that PosIR’s fine-grained position-aware relevance annotations are reliable and suitable for rigorous evaluation of position bias in retrieval models¹⁵.

C Multilingual Machine Translation

In this section, we detail the selection process of the translation model and the rigorous quality control protocols applied to construct the multilingual subset of PosIR.

Data Construction. To construct a representative evaluation set for translation quality, we perform stratified sampling from the 31 domain-specific English corpora utilized in PosIR. Within each domain, we select 12 documents stratified by token length into four bins (0–512, 512–1024, 1024–1536, and 1536–2048 tokens, measured by the Qwen3 tokenizer), yielding a total of 372 source texts. This design ensures balanced domain coverage while capturing a broad spectrum of document lengths and linguistic complexities.

Translation Models. We evaluate four representative translation systems: (1) *Google Translate*¹⁶, a widely used commercial neural machine translation service; (2) *Hunyuan-MT-7B* [38], a specialized translation model achieved first place in the WMT25 competition; (3) *Qwen3-30B-A3B-Instruct-2507* [29], a 31B-parameter open-source general MoE model; and (4) *GPT-4o* [19], a frontier general model serving as a strong baseline. Each system translates the sampled texts into 9 target languages: Arabic (ara-Arab), French (fra-Latn), German (deu-Latn), Italian (ita-Latn), Korean (kor-Kore),

Portuguese (por-Latn), Russian (rus-Cyrl), and Spanish (spa-Latn). The prompt for the translation task is provided in Appendix D.4.

C.1 Automatic Evaluation

C.1.1 Evaluation Protocol. To efficiently evaluate the translation quality across 8 languages and 31 domains, we employ an LLM-based reference-free evaluation approach [13] (see Appendix D.5). Specifically, we utilize GPT-4o as an automatic evaluator to assess translation quality along four dimensions: *Fluency* (naturalness and readability of the target text), *Accuracy* (semantic fidelity to the source), *Completeness* (preservation of all source information), and *Style Consistency* (appropriateness of tone and register). Each dimension is rated on a 5-point Likert scale (1 = Poor, 5 = Excellent), and an overall score is computed as the mean of the four dimensions. In total, this results in 2,976 evaluated translation outputs (372 samples × 8 languages).

C.1.2 Evaluation Results. Table 7 present aggregated evaluation results across four translation models and five quality dimensions. Qwen3-30B-A3B-Instruct-2507 achieves the highest overall score (4.54), followed by Google Translate (4.41), GPT-4o (4.36), and Hunyuan-MT-7B (3.89). Dimension-wise analysis reveals that Qwen3-30B-A3B-Instruct-2507 ranks first across all metrics, demonstrating particular strengths in *fluency* (4.65) and *style consistency* (4.63), while maintaining competitive performance in *accuracy* (4.44) and *completeness* (4.43). Google Translate exhibits a balanced profile with relatively uniform scores across dimensions, indicating stable but less exceptional translation quality. In stark contrast, Hunyuan-MT-7B shows severe degradation in *completeness* (3.28) and *accuracy* (3.72), with performance gaps exceeding 1.0 point compared to top-performing models. Despite achieving comparable *fluency*

¹⁵We acknowledge that a single annotator limits the assessment of inter-rater reliability; however, the near-ceiling scores across all dimensions suggest that annotation quality is unambiguous in the vast majority of cases.

¹⁶<https://translate.google.com>

Table 5: Summary of corpus statistics aggregated by domain.

Domain	Document Count			Token Length					
	Corpus	Queries	Ratio	Min	Q1	Median	Q3	Max	Mean
Accommodation Catering Hotel	439,993	14,612	2.56%	10	607	1173	1886	4432	1294.6
Aerospace	581,933	13,546	3.36%	9	610	1207	1955	4585	1326.1
Agriculture Forestry Animal Husbandry Fishery	581,329	16,175	3.37%	9	589	1231	2013	4719	1346.3
Artificial Intelligence Machine Learning	551,318	11,154	3.17%	5	538	1113	1884	4540	1276.7
Automobile	565,317	13,047	3.26%	5	584	1120	1870	4346	1271.3
Biomedicine	616,910	13,948	3.55%	7	580	1195	1972	4709	1327.3
Computer Communication	620,254	14,144	3.57%	6	607	1172	1908	4544	1300.5
Computer Programming Code	519,815	6,522	2.97%	3	701	1201	1853	4127	1307.0
Current Affairs Government Administration	606,998	12,832	3.49%	2	619	1225	1985	4785	1341.8
Electric Power Energy	599,389	13,119	3.45%	6	613	1220	2001	4676	1352.1
Film Entertainment	596,575	16,911	3.46%	10	592	1195	1906	4605	1295.0
Finance Economics	607,364	12,948	3.49%	5	626	1227	1982	4571	1343.5
Fineweb	698,633	16,748	4.03%	5	575	1175	1919	4645	1301.7
Fire Safety Food Safety	313,846	12,852	1.84%	13	672	1197	1953	4612	1361.1
Game	442,997	13,014	2.57%	6	646	1234	1970	4634	1339.5
Law Judiciary	603,544	12,947	3.47%	10	626	1221	1971	4676	1338.7
Literature Emotion	598,538	15,717	3.46%	3	613	1181	1915	4653	1299.6
Mathematics Statistics	618,864	10,324	3.54%	2	588	1140	1827	4180	1244.3
Medicine Health Psychology Traditional Chinese Medicine	633,903	13,925	3.65%	14	609	1211	1987	4828	1342.9
Mining	485,528	13,345	2.81%	9	510	1114	1890	4771	1266.1
News Media	539,996	12,879	3.12%	5	588	1230	1964	4738	1326.3
Other Information Services Information Security	321,187	8,590	1.86%	13	669	1246	2026	4402	1384.2
Other Manufacturing	596,244	16,025	3.45%	6	594	1131	1884	4444	1283.5
Petrochemical	577,348	12,803	3.32%	6	570	1180	1962	4583	1316.6
Real Estate Construction	567,575	16,259	3.29%	13	594	1211	1963	4693	1328.5
Sports	566,912	13,787	3.27%	13	565	1163	1892	4531	1281.9
Subject Education Education	618,409	16,796	3.58%	9	625	1221	1975	4569	1339.0
Technology Scientific Research	614,697	14,198	3.54%	3	595	1195	1960	4592	1318.4
Tourism Geography	549,553	14,974	3.18%	5	560	1176	1927	4633	1295.9
Transportation	547,312	14,129	3.16%	10	573	1190	1965	4640	1323.3
Water Resources Ocean	547,392	13,438	3.16%	7	571	1194	1960	4563	1325.8
Total/Avg	17,329,673	421,708	100.00%	7	600	1189	1939	4581	1316.1

Table 6: Distribution of query types in the English and Chinese retrieval datasets.

Query Type	English (%)	Chinese (%)
WHAT	34.46	41.44
HOW	26.60	17.93
WHY	12.33	23.18
WHICH	5.53	1.75
WHEN	3.14	5.68
WHERE	2.96	1.86
WHO	2.67	5.25
CLAIM	0.42	0.03
OTHERS	11.88	2.90
Total	100.00	100.00

score (4.37), qualitative analysis reveals that Hunyuan-MT-7B frequently sacrifices semantic fidelity to preserve surface-level linguistic smoothness, resulting in content omissions and meaning

distortions, particularly when handling complex or information-dense source texts. Moreover, as shown in Figure 7, a cross-linguistic analysis reveals that Romance languages such as French achieve relatively higher translation quality, whereas Arabic and Korean exhibit comparatively lower performance.

C.2 Human Evaluation

Initial inspection using automatic evaluation suggests that Qwen3-30B-A3B-Instruct-2507 produces high-quality translations across PosIR, achieving competitive performance in comparison to other methods. However, given that translation quality appears strong, rigorous human validation is essential. While automatic evaluation can detect catastrophic failures, it may be insensitive to subtle but critical errors in high-quality translations, such as nuanced mistranslations, contextual inappropriateness, or domain-specific inaccuracies. Due to limited resources, we conduct focused human evaluation on three typologically diverse languages (Arabic, French, and Russian) to provide expert assessment of translation quality,

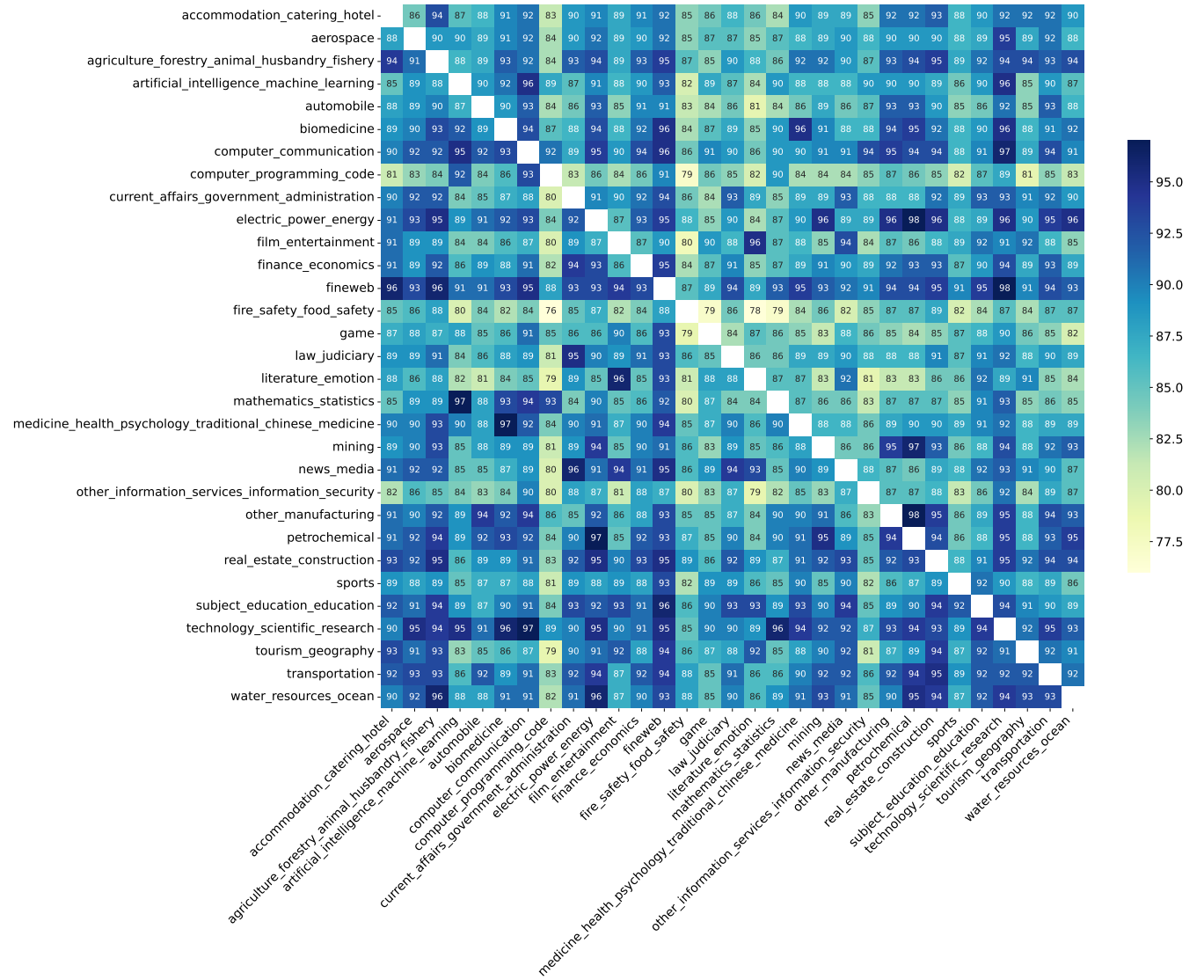


Figure 6: Inter-domain similarity heatmap. The lower triangular part of the matrix shows similarity scores computed from the English corpus, while the upper triangular part corresponds to those from the Chinese corpus. The diagonal is omitted for clarity.

facilitating a direct comparison between human evaluation and LLM-based assessments.

C.2.1 Annotator Recruitment. Five annotators were recruited from graduate programs in linguistics, natural language processing, and applied foreign languages via established academic networks. All annotators were advanced graduate students or recent PhD graduates with expertise in computational linguistics and/or translation studies. For Arabic and Russian, native speakers of the respective languages conducted the annotations. For French, two annotators holding DALF C1/C2 certificates (Diplôme Approfondi de Langue

Française, the highest level of French language certification) completed the evaluation. All annotators were compensated at a fair hourly rate, in compliance with applicable labor regulations.

C.2.2 Evaluation Protocol. Annotators evaluated 372 translation pairs across 3 languages (1,116 samples in total) generated by Qwen3-30B-A3B-Instruct-2507. The evaluations were conducted in accordance with the consistency standards outlined in the automatic evaluation, as described in Appendix C.1.1. Annotators were given both the English source text and the target language translation, with no access to reference translations to prevent anchoring bias.

Table 7: Automatic evaluation of multilingual translation quality by languages and dimensions (1-5 scale). Evaluation dimensions include Fluency (Flu), Accuracy (Acc), Completeness (Com), Style Consistency (Sty), and Overall Average Score (Ove). “Qwen3-30B-A3B” denotes the “Qwen3-30B-A3B-Instruct-2507” model.

Language	Qwen3-30B-A3B					Google Translate					Hunyuan-MT-7B					GPT-4o				
	Flu	Acc	Com	Sty	Ove	Flu	Acc	Com	Sty	Ove	Flu	Acc	Com	Sty	Ove	Flu	Acc	Com	Sty	Ove
Arabic	4.38	4.18	4.19	4.33	4.27	4.33	4.24	4.26	4.33	4.29	4.30	3.70	3.28	4.14	3.85	4.25	4.16	4.14	4.22	4.19
French	4.85	4.63	4.54	4.84	4.71	4.64	4.47	4.43	4.64	4.54	4.40	3.74	3.27	4.24	3.91	4.57	4.48	4.38	4.56	4.50
German	4.61	4.45	4.51	4.60	4.54	4.58	4.35	4.38	4.55	4.47	4.40	3.74	3.33	4.21	3.92	4.55	4.50	4.45	4.55	4.51
Italian	4.67	4.48	4.44	4.65	4.56	4.39	4.37	4.37	4.39	4.38	4.37	3.67	3.15	4.14	3.83	4.44	4.34	4.32	4.40	4.37
Korean	4.39	4.12	4.13	4.35	4.25	4.24	4.17	4.25	4.24	4.23	4.40	3.91	3.54	4.28	4.03	3.88	3.74	3.78	3.87	3.82
Portuguese	4.81	4.61	4.59	4.81	4.71	4.59	4.45	4.47	4.56	4.52	4.42	3.75	3.29	4.24	3.93	4.71	4.60	4.53	4.70	4.63
Russian	4.69	4.44	4.47	4.68	4.57	4.51	4.28	4.30	4.45	4.39	4.30	3.57	3.14	4.11	3.78	4.35	4.23	4.19	4.33	4.28
Spanish	4.77	4.60	4.61	4.77	4.69	4.52	4.39	4.40	4.52	4.46	4.37	3.71	3.22	4.22	3.88	4.63	4.52	4.49	4.62	4.57
Average	4.65	4.44	4.43	4.63	4.54	4.47	4.34	4.36	4.46	4.41	4.37	3.72	3.28	4.20	3.89	4.42	4.32	4.29	4.41	4.36

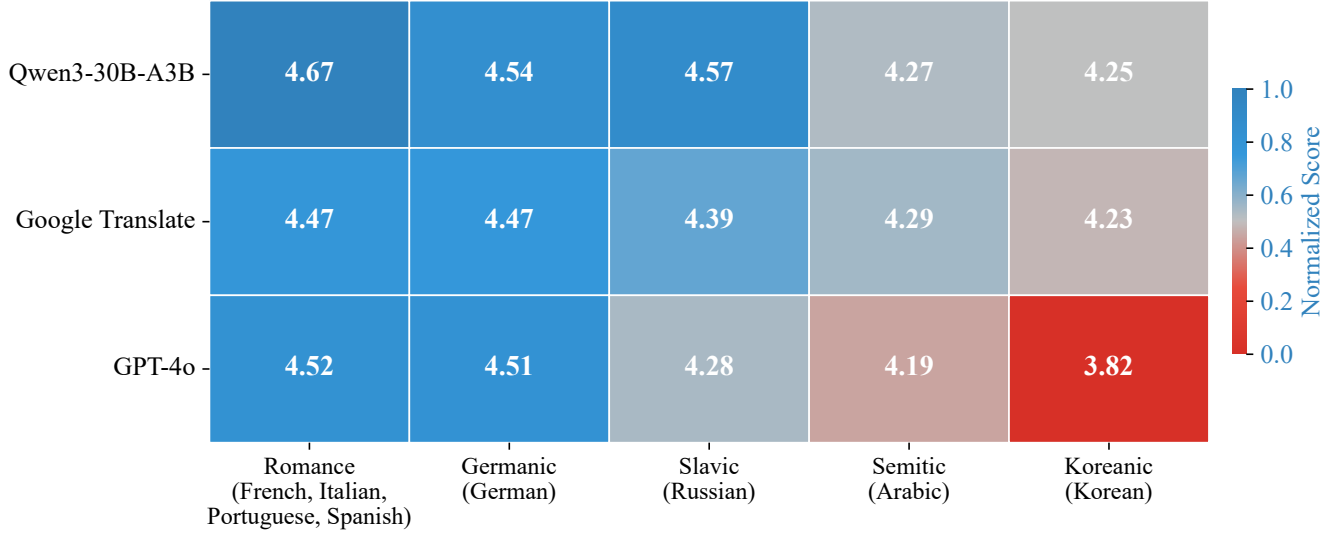


Figure 7: Cross-lingual translation quality heatmap (using Overall Average Score) across 3 translation models and 5 language families. “Qwen3-30B-A3B” denotes the “Qwen3-30B-A3B-Instruct-2507” model.

Table 8: Human evaluation scores by languages and dimensions (1-5 scale).

Dimension	French	Russian	Arabic	Avg.
Fluency	4.71	4.65	4.58	4.65
Accuracy	4.52	4.48	4.35	4.45
Completeness	4.89	4.82	4.76	4.82
Style Consistency	4.68	4.61	4.52	4.60
Overall Avg. Score	4.70	4.64	4.55	4.63

C.2.3 Evaluation Results. Table 8 presents the mean scores across languages and dimensions. All language-dimension combinations

achieved scores between 4.35 and 4.89, indicating consistently high translation quality. Over 97% of translations received scores ≥ 4 across all dimensions. Several patterns emerge from the results. First, *completeness* scores consistently achieve the highest ratings (4.76–4.89), indicating that Qwen3-30B-A3B-Instruct-2507 rarely omits source information and preserves semantic integrity well. Second, *fluency* scores consistently exceed *accuracy* scores across all languages (4.58–4.71 vs. 4.35–4.52), suggesting the model generates natural-sounding translations in target languages while maintaining acceptable fidelity. Third, language-specific differences reflect the typological distance from English: French achieves the highest average scores (4.70), likely due to abundant training data and linguistic proximity; Arabic shows relatively lower *accuracy* (4.35),

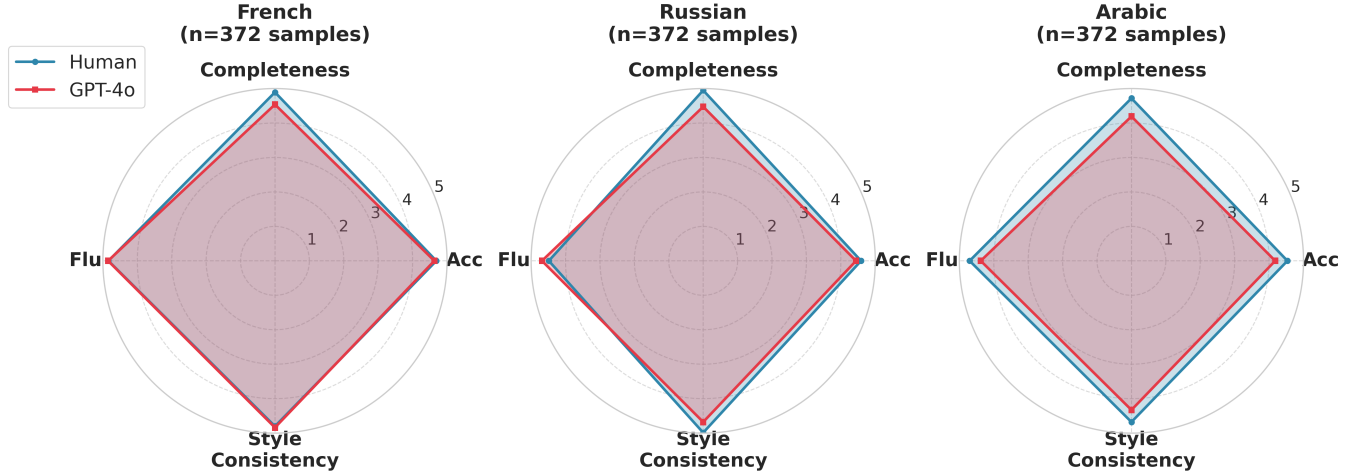


Figure 8: Human vs. GPT-4o score comparison by language. Evaluation dimensions include Fluency (Flu), Accuracy (Acc), Completeness, and Style Consistency. The overlapping radar shapes indicate strong alignment in quality assessment patterns.

Table 9: Human-GPT-4o agreement within ± 1 point tolerance.

Dimension	Agreement (± 1)	MAE
Fluency	98.1%	0.44
Accuracy	96.9%	0.51
Completeness	90.6%	0.57
Style Consistency	96.9%	0.39

potentially due to its complex morphological system; Russian performs in between (4.64), despite its challenging case system. The score distribution exhibits a pronounced ceiling effect: 67.1% of samples received 5 (Excellent), 30.4% received 4 (Good), and only 2.5% scored ≤ 3 . This confirms that the Qwen model produces high-quality translations suitable for PosIR, with $>97\%$ of samples rated Good or Excellent.

C.3 GPT-4o Human Evaluation Alignment

We validate GPT-4o as a translation evaluator by comparing its ratings against human scores on the same 1,116 samples. Figure 8 shows the score distribution comparison between human annotators and GPT-4o across languages. The radar charts demonstrate high shape alignment: both raters identify *completeness* as the strongest dimension and *accuracy* as relatively lower, with mean differences within 0.3 points across all dimensions. Under a ± 1 tolerance threshold (appropriate for ordinal Likert scales), GPT-4o achieves 96.15% agreement with human ratings. Table 9 shows per-dimension results. Figure 9 presents the detailed discrepancy analysis, revealing two notable patterns. In the high-score region, data concentrates along the 4–5 diagonal, explaining the strong agreement rates. However, for samples rated 5 by human annotators, GPT-4o averages 4.5–4.7, exhibiting a systematic offset of -0.3 to -0.5 points. This suggests GPT-4o acts as a stricter judge, more inclined to identify potential flaws than to endorse perfection, effectively serving as a conservative lower-bound estimator. Conversely,

in the sparsely populated low-score region, GPT-4o tends toward leniency: for the rare samples rated 1–3 by humans (only 2.5% of data), GPT-4o often assigns 4–5 scores, suggesting potential blind spots in detecting severe semantic errors. Fortunately, such low-quality samples are rare in our dataset, so GPT-4o’s conservative behavior on high-quality translations, where the vast majority of data resides, provides reliable quality assurance for PosIR. In summary, the strong human-GPT-4o alignment validates automated quality verification for the full dataset, establishing confidence in the translation reliability of the Qwen model.

C.4 Model Selection Rationale

Based on comprehensive evaluation results and practical considerations, we select *Qwen3-30B-A3B-Instruct-2507* as the primary translation model for constructing the PosIR multilingual corpus. This decision is justified by three key factors: (1) *superior translation quality*, consistently achieving the highest scores across all evaluation dimensions and target languages, with an overall advantage of 0.13 points over Google Translate and 0.21 points over GPT-4o; (2) *strong cross-lingual robustness*, maintaining stable performance across diverse linguistic typologies and writing systems without significant quality degradation; (3) *favorable resource efficiency*, delivering high translation quality with moderate computational requirements, which enables large-scale multilingual data construction under practical inference-time constraints. Additionally, as an open-source and freely accessible model, it offers further flexibility and cost-effectiveness for large-scale deployment and application.

C.5 Fixing and Filtering Strategies

Through manual inspection of the translated outputs, we observed recurring artifacts that could degrade data quality, such as invalid prefixes, refusal-style responses, and repetitive or verbose translations. To systematically mitigate these issues while preserving

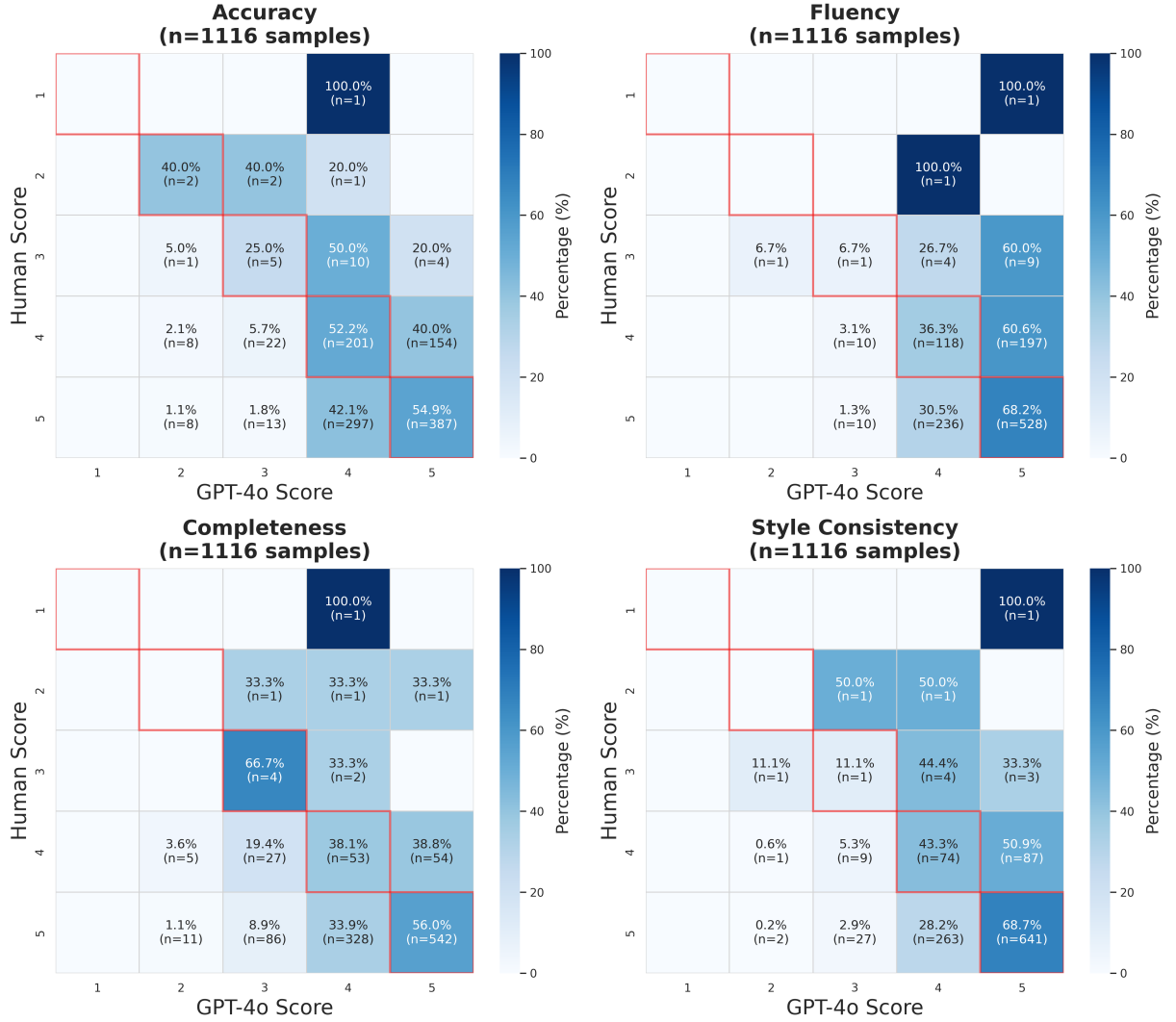


Figure 9: Confusion matrix of human vs. GPT-4o scores. Each row shows distribution of GPT-4o scores for a given human score. Data concentrates in the high-score diagonal (4–5), with GPT-4o showing systematic strictness for top-rated samples.

corpus-query-qrels consistency, we design and apply a three-stage cleaning pipeline.

C.5.1 Prefix Removal. The Qwen3-30B-A3B-Instruct-2507 model sometimes prepends explanatory text (e.g., “Here is the translation:”) to its outputs. To identify and remove such artifacts, we adopt a dual-strategy prefix detection heuristic. For corpus texts, we employ a co-occurrence pattern matching strategy that identifies the presence of translation-related keywords (“translate” or “translation”) and newline characters within a proximity window of 500 characters, starting from the first 500 characters and extending up to 2,000 characters. For queries, which are typically shorter, we apply full-text matching. Candidate prefixes are validated as English using langdetect¹⁷ and manually reviewed before compilation into a blacklist (118 patterns). Overall, this procedure affected

¹⁷<https://github.com/fedelopez77/langdetect>

54 instances across 40 files, accounting for 0.0003% of the 17.7M records and spanning 8 non-English languages.

C.5.2 Refusal Detection. We employ a two-stage filtering approach to identify translation refusals. In the first stage, we apply lightweight blacklist matching combined with language detection to flag candidate refusals, prioritizing high recall with minimal computational overhead. In the second stage, flagged instances are validated using Qwen3-Max¹⁸, which inspects the first 1,500 characters to distinguish refusal-related meta-commentary (e.g., “I cannot translate”) from legitimate translations. This process identified 2,249 refusals, predominantly in Arabic (1,378 instances) and Korean (323). We remove 2,223 corpus entries and 26 queries, with cascading deletion of 26 corresponding qrels to maintain data consistency.

¹⁸<https://qwen.ai/blog?id=qwen3-max>

C.5.3 Statistical Outlier Filtering. We perform token-length-based outlier filtering at the language-domain level using the Qwen3 tokenizer. For each stratum, we compute the interquartile range (IQR) and define asymmetric thresholds, with a lower bound of $Q1 - 3.0 \times IQR$ and an upper bound of $Q3 + 1.5 \times IQR$. This asymmetric design focuses on filtering abnormally long texts, which are more likely to result from translation errors, duplication, or unintended document concatenation. It enforces a stricter upper bound to remove such cases, while allowing reasonable length variation among shorter documents. We process only corpus outliers, cascading deletions to associated queries and qrels.

In total, 15,641 corpus documents (0.11%) and 332 queries (0.10%) are removed. The largest reductions occur in Arabic (5,490 corpus entries; 0.25%) and Korean (4,082; 0.24%), followed by Russian (1,557; 0.11%). Overall, the removal ratios remain consistently low across all languages, indicating that the filtering procedure effectively eliminates anomalous long texts while preserving the structural integrity and cross-lingual balance of the benchmark.

D Prompts

D.1 Prompt for Position-Aware QA Generation

We use the following prompt to instruct an LLM to generate diverse and realistic question-answer (QA) pairs under specific positional constraints. Unlike explicitly segmenting the document into segments, we always feed the *entire* document to the LLM. Position control is achieved solely by injecting a sampled positional constraint into a dedicated slot in the prompt.

Sampling and Constraint Injection. For each generation instance, we randomly sample one positional constraint from a predefined set and insert it into the prompt via **Positional Requirement Slot**. This constraint guides the LLM to produce questions whose answers primarily come from a particular region of the document, while the model still has access to the complete document context. The Positional Requirement Slot is filled with exactly one of the following options:

- *Focus on first third: The question should primarily reference the first third of the document.*
- *Focus on middle third: The question should primarily reference the middle third of the document.*
- *Focus on final third: The question should primarily reference the final third of the document.*

Prompt Usage. The prompt below enforces key requirements including answerability, naturalness, and the avoidance of meta references to the source document. The positional constraint is an *internal control signal* and must not be mentioned explicitly in the generated questions. The question type list is intentionally over-complete to support diverse generation; not all types are expected to be instantiated for every document.

Task Description

Generate user questions in realistic search engine scenarios based on a given document.

Core Requirements

1. **Answerability:** Answers must be explicitly stated or directly inferable from the document (no outside knowledge).
2. **Authenticity:** Questions must reflect natural human phrasing.
3. **No Referencing:** Avoid terms like “this text”, “the author”, or “this document”.
4. **{{Positional Requirement Slot}}**

Question Configuration

Configure two elements: Question Type and Expression Style. Select configurations contextually.

Question Type

1. ****Fact Query**:** Retrieve specific information (e.g., dates, metrics, definitions).
2. ****Operation Guide**:** Request steps/methods to accomplish tasks (focuses on actionable steps).
3. ****Cause Analysis**:** Investigate reasons behind occurrences or phenomena.
4. ****Comparison Choice**:** Contrast differences/advantages to facilitate decisions.
5. ****Concept Explanation**:** Clarify meanings, categories, or structures of terms/concepts.
6. ****Viewpoint Verification**:** Validate accuracy/reliability of claims/data (requires credibility evaluation).
7. ****Prediction/Speculation**:** Inquire about future outcomes or plausible inferences strictly grounded in the document content.
8. ****True/False Judgment**:** Questions answerable with “yes” or “no”.
9. ****Background Exploration**:** Understand context, history, current status, or related environments.
10. ****Seeking Advice**:** Request subjective opinions, experiences, or recommendations (emphasizes actionable suggestions).
11. ****Open Discussion**:** Pose broad/debatable topics to stimulate dialogue.
12. ****Emotional Resonance**:** Express feelings (frustration/seeking help/excitement) to obtain empathy/support.
13. ****Feasibility Assessment**:** Query viability, costs, or resource requirements for plans/ideas.
14. ****Mathematical Reasoning**:** Problems needing numerical calculations or logical deductions (require step-by-step solutions).
15. ****Content Creation**:** Request original text, narratives, or creative output.
16. ****Relationship Counseling**:** Focuses on conflict resolution in interpersonal interactions, communication strategies, or emotional advice.
17. ****Diagnostic Troubleshooting**:** Analyzes the root causes of issues based on anomalies and provides solutions.
18. ****Hypothetical Scenarios**:** Explores possibilities,

ethical implications, or logical inferences under fictional or extreme conditions.

19. **Decision Advisory**: Assists in analyzing pros and cons, weighing factors for significant or personalized choices.

20. **Personal Growth**: Requests specific strategies, methods, or learning path planning for skill development, habit formation, efficiency enhancement, goal setting, or mindset adjustment.

Expression Style

- Concise | Casual | Informal
- Formal | Professional
- Technical | Academic

Steps

Follow the steps below and output results for each step in order.

1. Scenario Setup: Envision potential users and contexts where this document would be relevant.
2. Type Selection: Identify appropriate question types based on the content emphasized by the sampled positional constraint.
3. Style Matching: Select compatible expression styles.
4. Final Generation: Generate the final QA pairs and present them as a JSON array enclosed in a Markdown code block.

```
[
  {
    "question": "Generated question",
    "answer": "Answer text"
  }
]
```

Important Tips

1. Users do not know the document content. Questions must (a) strictly avoid document references, and (b) use natural phrasing.
2. The positional constraint is for generation control only; do not reveal or mention it in the questions.
3. For subjective question types (e.g., advice), answers must be grounded in the document's stated information; do not introduce external opinions.
4. If the document quality prevents question generation, output: "Document quality insufficient for generation."

Now, let's start this task:

```
<document>{{DOCUMENT}}</document>
```

D.2 Prompt for Reference Locating

Although explicit positional constraints are imposed during QA generation, the resulting questions may still violate these constraints in practice. To address this issue, we design the following prompt to instruct an LLM to locate supporting reference text within the original document, thereby enhancing the factual grounding and answerability of the generated questions.

Task Description

Given a list of questions and a document, for each question, **precisely identify and extract continuous text segments** from the document that **directly answer or support answering** this question.

Requirements

1. **Text Integrity**: Copy document content verbatim. Do not paraphrase, summarize, or splice across paragraphs. Preserve original punctuation, whitespace, and line breaks exactly.
2. **Extraction Principles**: Prioritize complete clauses/semantic units as minimal extraction units; If multiple relevant segments exist, output all segments completely without omission.
3. **Relevance Criteria**: Extracted segments must satisfy **at least one** of: (a) Containing direct answers to the question; (b) Providing essential contextual support for answering the question
4. **Question Coverage**: Process **every question** in the provided question list.
5. **Accurately Assess Question Answerability**: If the document can answer the question (directly or indirectly), extract the reference text as supporting evidence; If the document cannot provide useful information, return an empty list.

Output Specification: Use strictly this JSON format.

```
[
  {
    "question": "Original Question",
    "reference_texts": [
      "Document Segment 1",
      "Document Segment 2",
      ...
    ]
  }
]
```

Now, let's start this task:

```
<document>{{DOCUMENT}}</document>
<questions>{{QUESTIONS}}</questions>
```

D.3 Prompt for Relevance Assessment

Previous work [27] demonstrate that large language models such as GPT-4 can achieve labeling quality comparable to human annotators when producing gold-standard relevance labels for search systems. Moreover, [33] shows that incorporating fine-grained relevance labels into prompts for LLM-based rerankers significantly improves zero-shot reranking performance by better distinguishing borderline and ambiguous relevance cases. Motivated by these findings, we employ DeepSeek-V3.1 (Non-thinking Mode)¹⁹ as a relevance labeler, adopting a 5-level relevance labeling scheme to

¹⁹<https://api-docs.deepseek.com/news/news250821>

capture fine-grained distinctions in relevance. This labeling process is used solely for auxiliary verification rather than primary relevance annotation.

Task

Given a list of questions and a document, evaluate the relevance between each question and the document on a 0-4 scale:

0 = Completely irrelevant

1 = Slightly relevant (minimal connection)

2 = Moderately relevant (The document addresses parts of the question but lacks completeness, depth, or direct alignment)

3 = Highly relevant (covers most aspects of the question)

4 = Perfectly relevant (can correctly address the question)

Output Format

Output all the relevance scores in the **same order** as the input questions in the format of a JSON array in Markdown. An example, assuming there are three questions, then your output is:

```
[
  1,
  3,
  0
]
```

Now, let's start this task:

```
<document>{{DOCUMENT}}</document>
<questions>{{QUESTIONS}}</questions>
```

D.4 Prompt for Multilingual Machine Translation

You are a world-class professional translator, fluent in both English and {{LANGUAGE}}.

Your task is to translate the following text into {{LANGUAGE}}.

Please adhere to the highest standards of translation, ensuring the output is:

1. **Faithful**: Accurately convey the original meaning, content, and intent without omission or distortion.
2. **Expressive**: Make the translation smooth, natural, and easy to understand for a native speaker of the {{LANGUAGE}}.
3. **Elegant**: Preserve the style, tone, and literary grace of the original text, making it aesthetically pleasing to read.

Here is the text:

```
{{TEXT}}
```

D.5 Prompt for Translation Quality Evaluation

Task

You are a professional translation quality evaluator. Evaluate the English to {{LANGUAGE}} translation using a 5-point integer scale (1-5, whole numbers only).

Evaluation Criteria

1. **Accuracy**: Semantic fidelity to original meaning - no mistranslations, omissions, or additions.
2. **Fluency**: Natural language quality, grammar, and readability in target language.
3. **Completeness**: All source content translated without missing information or concepts.
4. **Style Consistency**: Appropriate tone, register, and style matching the source text context.

Scoring Scale

- **5**: Excellent (professional publication quality, no issues).
- **4**: Good (minor issues that don't affect understanding).
- **3**: Acceptable (noticeable issues but meaning is clear).
- **2**: Poor (significant problems affecting comprehension).
- **1**: Unacceptable (major errors, meaning unclear or wrong).

Output Format

Provide ONLY this JSON format (no additional text):

```
{
  "accuracy": <1-5>,
  "fluency": <1-5>,
  "completeness": <1-5>,
  "style_consistency": <1-5>,
  "comments": "Brief justification"
}
```

Now, let's start this task:

Input

Original Text (English)

```
{{ORIGINAL_TEXT}}
```

Translation ({{LANGUAGE}})

```
{{TRANSLATED_TEXT}}
```

Table 10: Detailed evaluation results of 10 multilingual retrieval models on PosIR (Part 1 of 2). In both multilingual retrieval and cross-lingual retrieval (translated queries retrieving English documents) settings, the results for each language are weighted-averaged across 31 domains. Q1–Q4 represent query buckets partitioned by the token length of positive documents (512-token intervals). “KaLM-Embedding-12B” denotes the “KaLM-Embedding-Gemma3-12B-2511” model. For Part 2, refer to Table 11.

	Model	Arabic		Chinese		German		English		French		Italian		Korean		Portuguese		Russian		Spanish		
		nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	
PosIR	Multilingual Retrieval																					
	gte-multilingual-base	34.08	0.52	51.52	0.26	47.96	0.44	61.59	0.19	49.59	0.41	47.63	0.45	37.54	0.48	48.94	0.42	44.89	0.46	49.95	0.42	
	bge-m3	34.08	0.41	47.93	0.34	46.59	0.29	50.79	0.30	44.07	0.33	42.50	0.34	38.19	0.35	44.27	0.36	40.60	0.36	43.16	0.35	
	Qwen3-Embedding-0.6B	42.11	0.45	57.49	0.30	53.79	0.37	65.10	0.22	55.33	0.34	54.49	0.36	45.68	0.38	55.68	0.35	50.20	0.40	56.43	0.33	
	inf-retriever-v1-1.5b	46.48	0.32	64.12	0.17	58.83	0.23	69.26	0.09	60.85	0.21	59.39	0.25	51.52	0.25	60.94	0.23	55.49	0.23	61.21	0.23	
	Qwen3-Embedding-4B	54.10	0.32	60.32	0.31	63.67	0.25	69.96	0.22	62.47	0.27	63.84	0.26	57.26	0.27	64.78	0.26	60.51	0.30	65.67	0.25	
	inf-retriever-v1	55.03	0.13	68.58	0.15	65.59	0.10	72.78	0.08	66.47	0.11	65.76	0.10	58.65	0.13	67.01	0.11	63.02	0.08	67.18	0.10	
	NV-Embed-v2	18.32	0.75	37.46	0.56	51.28	0.49	68.27	0.32	52.40	0.48	50.79	0.50	28.17	0.72	52.41	0.51	37.42	0.62	53.67	0.49	
	llama-embed-nemotron-8b	55.41	0.07	60.01	0.11	66.34	0.06	73.48	0.06	65.73	0.05	65.79	0.06	58.78	0.07	66.67	0.07	61.94	0.06	66.71	0.06	
	Qwen3-Embedding-8B	57.43	0.20	60.36	0.32	65.30	0.18	70.27	0.20	65.32	0.15	65.70	0.18	59.57	0.17	66.57	0.20	62.95	0.19	67.32	0.19	
	KaLM-Embedding-12B	43.74	0.11	50.82	0.11	53.33	0.08	64.22	0.06	52.40	0.06	52.15	0.07	43.80	0.10	51.71	0.07	51.73	0.08	54.79	0.06	
	Cross-lingual Retrieval																					
gte-multilingual-base	31.28	0.28	-	-	50.75	0.26	-	-	53.47	0.24	51.62	0.27	32.88	0.23	53.38	0.24	44.45	0.28	54.06	0.24		
bge-m3	24.04	0.39	-	-	43.16	0.35	-	-	41.70	0.35	40.37	0.35	28.06	0.43	41.26	0.36	31.83	0.39	39.97	0.36		
Qwen3-Embedding-0.6B	34.91	0.38	-	-	53.97	0.29	-	-	55.14	0.28	54.28	0.29	39.50	0.32	55.52	0.28	47.68	0.29	56.30	0.27		
inf-retriever-v1-1.5b	35.68	0.23	-	-	57.33	0.14	-	-	60.08	0.12	58.19	0.13	42.74	0.18	60.14	0.13	50.34	0.17	60.39	0.13		
Qwen3-Embedding-4B	52.20	0.32	-	-	63.38	0.25	-	-	62.88	0.27	63.49	0.26	55.74	0.28	64.09	0.27	59.50	0.28	64.80	0.26		
inf-retriever-v1	52.28	0.17	-	-	65.34	0.10	-	-	66.77	0.10	65.89	0.10	56.62	0.14	66.94	0.10	61.82	0.12	67.28	0.10		
NV-Embed-v2	20.60	0.34	-	-	56.47	0.30	-	-	58.42	0.30	57.20	0.30	20.29	0.39	58.75	0.30	37.48	0.34	58.52	0.30		
llama-embed-nemotron-8b	52.73	0.08	-	-	65.80	0.04	-	-	66.18	0.06	66.21	0.05	56.35	0.08	66.74	0.06	60.47	0.05	66.94	0.05		
Qwen3-Embedding-8B	53.97	0.26	-	-	64.33	0.22	-	-	64.47	0.22	64.20	0.23	57.28	0.25	64.92	0.23	60.81	0.24	65.47	0.22		
KaLM-Embedding-12B	47.61	0.11	-	-	57.65	0.07	-	-	58.18	0.07	57.85	0.07	50.90	0.10	58.23	0.07	54.92	0.08	58.54	0.07		
Q1(512)	Multilingual Retrieval																					
	gte-multilingual-base	49.55	0.33	61.52	0.22	61.47	0.21	73.24	0.07	64.12	0.20	61.56	0.22	54.06	0.28	62.82	0.19	59.86	0.24	64.62	0.18	
	bge-m3	46.29	0.40	62.37	0.34	59.89	0.21	66.00	0.23	58.53	0.28	56.63	0.29	52.32	0.32	58.36	0.28	53.41	0.31	57.82	0.29	
	Qwen3-Embedding-0.6B	50.25	0.35	69.10	0.19	62.12	0.21	77.34	0.10	63.80	0.21	62.89	0.18	54.66	0.23	64.43	0.21	59.04	0.29	65.70	0.17	
	inf-retriever-v1-1.5b	55.72	0.32	71.77	0.16	67.87	0.22	77.89	0.07	69.96	0.18	68.72	0.22	60.70	0.24	70.10	0.21	64.80	0.23	70.66	0.18	
	Qwen3-Embedding-4B	63.67	0.18	73.96	0.17	71.79	0.13	81.54	0.11	70.39	0.12	71.76	0.12	66.46	0.14	72.93	0.12	69.65	0.13	74.16	0.09	
	inf-retriever-v1	65.89	0.19	75.98	0.14	75.03	0.14	81.09	0.07	76.27	0.13	76.08	0.16	70.56	0.20	76.73	0.11	72.39	0.13	77.08	0.12	
	NV-Embed-v2	43.77	0.54	62.36	0.20	77.81	0.12	80.81	0.08	77.95	0.11	78.12	0.10	59.23	0.37	79.26	0.09	66.76	0.16	78.72	0.09	
	llama-embed-nemotron-8b	67.74	0.21	71.37	0.24	77.14	0.11	84.68	0.06	77.09	0.11	77.15	0.11	72.35	0.19	77.93	0.11	73.97	0.14	78.13	0.11	
	Qwen3-Embedding-8B	64.85	0.15	74.56	0.19	72.10	0.12	82.50	0.12	72.20	0.11	73.31	0.12	67.32	0.12	74.38	0.09	70.31	0.13	75.27	0.08	
	KaLM-Embedding-12B	68.66	0.18	66.89	0.12	76.11	0.12	79.16	0.07	75.71	0.10	75.72	0.11	70.58	0.16	75.76	0.09	74.48	0.10	77.04	0.07	
	Cross-lingual Retrieval																					
gte-multilingual-base	41.11	0.25	-	-	60.45	0.14	-	-	63.98	0.11	61.67	0.13	44.02	0.23	64.09	0.10	55.56	0.18	64.84	0.11		
bge-m3	35.34	0.37	-	-	54.93	0.31	-	-	55.23	0.30	53.57	0.31	40.79	0.41	54.91	0.31	44.64	0.38	54.00	0.29		
Qwen3-Embedding-0.6B	45.85	0.23	-	-	65.33	0.11	-	-	66.56	0.10	65.90	0.13	51.56	0.19	67.32	0.11	59.87	0.16	68.43	0.09		
inf-retriever-v1-1.5b	43.60	0.30	-	-	64.51	0.17	-	-	68.19	0.13	66.04	0.16	51.07	0.26	68.50	0.14	59.11	0.18	68.84	0.12		
Qwen3-Embedding-4B	65.11	0.14	-	-	75.41	0.14	-	-	74.92	0.12	75.53	0.13	69.10	0.13	76.11	0.13	72.29	0.12	76.99	0.12		
inf-retriever-v1	57.93	0.20	-	-	71.70	0.13	-	-	73.67	0.11	72.51	0.11	63.78	0.16	73.74	0.09	68.28	0.14	74.21	0.10		
NV-Embed-v2	20.78	0.33	-	-	64.61	0.11	-	-	68.83	0.13	65.27	0.12	23.73	0.26	69.30	0.10	41.04	0.16	67.86	0.09		
llama-embed-nemotron-8b	65.40	0.15	-	-	76.99	0.08	-	-	77.66	0.09	77.59	0.09	70.22	0.15	78.29	0.07	73.22	0.12	78.62	0.07		
Qwen3-Embedding-8B	68.63	0.12	-	-	77.43	0.13	-	-	77.57	0.12	77.53	0.12	72.25	0.13	78.27	0.12	74.80	0.12	78.73	0.12		
KaLM-Embedding-12B	66.68	0.15	-	-	74.57	0.11	-	-	75.01	0.09	74.95	0.09	68.85	0.12	75.43	0.08	72.79	0.09	75.30	0.07		
Q2(1024)	Multilingual Retrieval																					
	gte-multilingual-base	35.75	0.58	52.22	0.28	49.87	0.48	62.56	0.24	50.17	0.46	49.25	0.47	39.15	0.49	50.84	0.47	46.55	0.51	51.54	0.46	
	bge-m3	34.85	0.51	46.42	0.37	47.28	0.40	51.56	0.37	44.15	0.43	43.06	0.44	37.91	0.43	44.71	0.45	41.04	0.46	43.80	0.44	
	Qwen3-Embedding-0.6B	43.35	0.48	56.50	0.36	54.98	0.37	65.69	0.29	56.37	0.36	55.52	0.39	46.30	0.41	56.61	0.37	51.47	0.43	57.29	0.38	
	inf-retriever-v1-1.5b	46.65	0.39	62.60	0.22	58.51	0.30	69.11	0.17	60.21	0.27	58.98	0.29	50.97	0.30	60.83	0.30	55.46	0.28	60.68	0.26	
	Qwen3-Embedding-4B	55.19	0.36	59.42	0.32	64.72	0.28	70.65	0.24	63.50	0.29	64.69	0.28	57.69	0.30	65.58	0.29	61.23	0.31	66.40	0.28	
	inf-retriever-v1	55.56	0.25	67.03	0.22	65.75	0.21	72.93	0.16	66.37	0.21	66.08	0.19	58.47	0.25	67.28	0.22	63.62	0.19	67.24	0.21	
	NV-Embed-v2	11.80	0.84	41.32	0.61	54.51	0.47	72.56	0.11	57.53	0.43	55.16	0.53	24.85	0.77	56.90	0.47	37.44	0.67	59.14	0.43	
	llama-embed-nemotron-8b	55.91	0.22	59.46	0.20	66.67	0.16	73.56	0.14	65.87	0.18	65.97	0.16	58.45	0.19	66.80	0.18	62.00	0.17	66.88	0.16	
	Qwen3-Embedding-8B	58.28	0.26	59.76	0.31	65.66	0.23	71.31	0.21	65.61	0.19	65.96	0.21	59.36	0.21	66.67	0.21	63.13	0.25	67.58	0.21	
	KaLM-Embedding-12B	43.63	0.33	54.69	0.21	56.04	0.25	71.77	0.17	54.72	0.26	54.35	0.25	44.52	0.33	55.23	0.27	52.84	0.26	58.59	0.25	
	Cross-lingual Retrieval																					
gte-multilingual-base	31.88	0.41	-	-	51.87	0.29	-	-	54.34	0.28	52.38	0.30	33.13	0.39	54.10	0.27	45.32	0.34	54.80	0.30		
bge-m3	23.61	0.56	-	-	43.68	0.43	-	-	41.91	0.42	40.63	0.46	27.17	0.51	41.57	0.42	31.54	0.53	40.26	0.45		
Qwen3-Embedding-0.6B	34.76	0.47	-	-	54.07	0.35	-	-	55.18	0.35	54.22	0.33	39.07	0.41	55.64	0.34	47.54	0.40	56.29	0.33		
inf-retriever-v1-1.5b	34.80	0.37	-	-	57.08	0.26	-	-	59.37	0.23	57.53	0.24	42.48	0.29	59.63	0.23	49.28	0.27	59.79	0.22		
Qwen3-Embedding-4B	52.24	0.34	-	-	63.89	0.29	-	-	62.95	0.30	63.62	0.30	55.76	0.33	64.16	0.29	59.46	0.31	65.05	0.28		
inf-retriever-v1	52.74	0.28	-	-	65.53	0.19	-	-	66.86	0.20	66.05	0.21	56.76	0.24	67.24	0.21	62.12	0.23	67.37	0.22		
NV-Embed-v2	24.07	0.22	-	-	60.85	0.13	-	-	62.60	0.11	62.05	0.13	22.51	0.15	63.08	0.11	42.00	0.16	62.89	0.12		
llama-embed-nemotron-8b	52.25	0.26	-	-	65.85	0.18	-	-	66.03	0.17	65.97	0.18	55.72	0.22	66.50	0.18	59.91	0.23	66.72	0.17		
Qwen3-Embedding-8B	54.79	0.28	-	-	65.10	0.25	-	-	65.02	0.26	64.63											

Table 11: Detailed evaluation results of 10 multilingual retrieval models on PosIR (Part 2 of 2). In both multilingual retrieval and cross-lingual retrieval (translated queries retrieving English documents) settings, the results for each language are weighted-averaged across 31 domains. Q1–Q4 represent query buckets partitioned by the token length of positive documents (512-token intervals). “KaLM-Embedding-12B” denotes the “KaLM-Embedding-Gemma3-12B-2511” model.

Model	Arabic		Chinese		German		English		French		Italian		Korean		Portuguese		Russian		Spanish	
	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI	nDCG@10	PSI
<i>Multilingual Retrieval</i>																				
gte-multilingual-base	18.41	0.78	39.21	0.40	32.10	0.67	48.46	0.38	35.11	0.62	31.76	0.67	20.19	0.74	32.57	0.64	28.78	0.69	33.50	0.63
bge-m3	22.82	0.48	34.22	0.42	32.62	0.42	34.30	0.40	30.98	0.40	28.76	0.46	25.69	0.44	30.57	0.47	28.17	0.42	29.08	0.46
Qwen3-Embedding-0.6B	34.18	0.57	43.84	0.44	44.13	0.51	50.39	0.39	45.78	0.47	45.08	0.49	37.03	0.52	45.55	0.48	41.43	0.53	45.79	0.49
inf-retriever-v1-1.5b	39.18	0.36	57.73	0.14	51.06	0.30	61.13	0.16	53.37	0.26	51.65	0.30	44.03	0.32	52.83	0.28	47.65	0.26	53.40	0.31
Qwen3-Embedding-4B	43.76	0.48	43.94	0.49	53.04	0.41	54.94	0.41	52.09	0.43	54.05	0.42	47.61	0.47	54.40	0.40	50.46	0.45	55.29	0.39
inf-retriever-v1	43.55	0.20	63.03	0.17	55.31	0.16	63.61	0.15	56.13	0.20	54.21	0.16	46.81	0.20	55.77	0.17	52.39	0.20	55.99	0.16
NV-Embed-v2	0.21	1.00	9.78	0.92	20.25	0.74	47.37	0.57	18.90	0.74	16.97	0.74	0.66	1.00	19.00	0.75	9.18	0.85	20.42	0.75
llama-embed-nemotron-8b	43.13	0.27	49.47	0.28	54.75	0.21	61.77	0.15	54.05	0.20	54.11	0.21	45.84	0.24	55.01	0.19	49.57	0.25	55.13	0.20
Qwen3-Embedding-8B	48.73	0.33	43.69	0.48	56.83	0.30	54.68	0.40	56.97	0.27	56.64	0.30	51.44	0.30	57.14	0.32	54.86	0.32	57.76	0.33
KaLM-Embedding-12B	20.88	0.40	30.88	0.29	27.99	0.29	41.33	0.17	27.51	0.30	26.83	0.36	18.55	0.37	24.85	0.35	28.67	0.34	29.00	0.30
<i>Cross-lingual Retrieval</i>																				
gte-multilingual-base	21.80	0.49	-	-	39.50	0.41	-	-	41.98	0.41	40.45	0.43	22.27	0.47	41.37	0.40	32.99	0.43	42.19	0.40
bge-m3	14.47	0.53	-	-	31.06	0.44	-	-	28.02	0.42	26.87	0.46	17.50	0.53	26.98	0.45	19.85	0.50	25.76	0.45
Qwen3-Embedding-0.6B	23.99	0.61	-	-	41.08	0.48	-	-	42.09	0.45	41.08	0.47	27.83	0.59	41.76	0.48	34.97	0.48	42.65	0.50
inf-retriever-v1-1.5b	30.48	0.24	-	-	50.96	0.18	-	-	53.05	0.14	51.61	0.13	36.60	0.27	52.93	0.17	43.84	0.20	53.06	0.19
Qwen3-Embedding-4B	37.33	0.50	-	-	48.60	0.44	-	-	48.43	0.45	48.86	0.44	40.74	0.52	49.31	0.46	44.62	0.48	49.84	0.44
inf-retriever-v1	46.25	0.18	-	-	58.48	0.16	-	-	59.01	0.16	58.27	0.16	49.89	0.19	58.80	0.13	54.42	0.16	59.36	0.14
NV-Embed-v2	16.81	0.66	-	-	41.08	0.59	-	-	40.98	0.60	40.78	0.59	14.19	0.70	40.48	0.59	27.57	0.64	41.57	0.59
llama-embed-nemotron-8b	41.60	0.19	-	-	54.61	0.18	-	-	54.99	0.17	55.18	0.17	44.35	0.20	55.45	0.17	48.75	0.19	55.58	0.20
Qwen3-Embedding-8B	37.03	0.47	-	-	48.14	0.44	-	-	48.58	0.42	48.04	0.45	40.78	0.48	48.71	0.45	44.48	0.45	49.36	0.45
KaLM-Embedding-12B	20.69	0.26	-	-	32.23	0.22	-	-	32.94	0.21	32.26	0.25	25.30	0.26	32.42	0.23	28.67	0.24	33.43	0.22

Table 12: Detailed information on all of the models appearing in our paper.

Model	Size	Open-Sourced	Model Link
<i>Tokenizer</i>			
Qwen3 tokenizer [29]	-	✓	https://huggingface.co/Qwen/Qwen3-30B-A3B-Instruct-2507/blob/main/tokenizer.json
<i>Embedding Model</i>			
gte-multilingual-base [35]	305M	✓	https://huggingface.co/Alibaba-NLP/gte-multilingual-base
bge-m3 [3]	568M	✓	https://huggingface.co/BAAI/bge-m3
Qwen3-Embedding-0.6B [36]	595M	✓	https://huggingface.co/Qwen/Qwen3-Embedding-0.6B
inf-retriever-v1-1.5b	2B	✓	https://huggingface.co/infly/inf-retriever-v1-1.5b
Qwen3-Embedding-4B [36]	4B	✓	https://huggingface.co/Qwen/Qwen3-Embedding-4B
inf-retriever-v1	7B	✓	https://huggingface.co/infly/inf-retriever-v1
QZhou-Embedding [30]	7B	✓	https://huggingface.co/Kingsoft-LLM/QZhou-Embedding
NV-Embed-v2 [14]	8B	✓	https://huggingface.co/nvidia/NV-Embed-v2
llama-embed-nemotron-8b [1]	8B	✓	https://huggingface.co/nvidia/llama-embed-nemotron-8b
Qwen3-Embedding-8B [36]	8B	✓	https://huggingface.co/Qwen/Qwen3-Embedding-8B
KaLM-Embedding-Gemma3-12B-2511 [37]	12B	✓	https://huggingface.co/tencent/KaLM-Embedding-Gemma3-12B-2511
<i>Re-ranking Model</i>			
bge-reranker-v2-m3 [3]	0.6B	✓	https://huggingface.co/BAAI/bge-reranker-v2-m3
Qwen3-Reranker-0.6B [36]	0.6B	✓	https://huggingface.co/Qwen/Qwen3-Reranker-0.6B
bge-reranker-v2-minicpm-layerwise [3]	3B	✓	https://huggingface.co/BAAI/bge-reranker-v2-minicpm-layerwise
Qwen3-Reranker-4B [36]	4B	✓	https://huggingface.co/Qwen/Qwen3-Reranker-4B
<i>Large Language Model</i>			
Hunyuan-MT-7B [38]	8B	✓	https://huggingface.co/tencent/Hunyuan-MT-7B
Qwen3-30B-A3B-Instruct-2507 [29]	31B	✓	https://huggingface.co/Qwen/Qwen3-30B-A3B-Instruct-2507
DeepSeek-V3.1 [7]	685B	✓	https://huggingface.co/deepseek-ai/DeepSeek-V3.1
GPT-4o [19]	-	✗	https://platform.openai.com/docs/models/gpt-4o
<i>Online Service</i>			
Google Translate	-	✗	https://translate.google.com

Table 13: Statistics for domain: Accommodation Catering Hotel (Part 1 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	42,596	1511.2	166.0%	1458	24.4	193.6%
Chinese	55,969	1009.2	-	1488	11.8	—
German	42,690	1339.6	147.1%	1459	20.7	164.2%
English	42,708	910.6	-	1459	12.6	—
French	42,699	1342.6	147.4%	1459	22.0	174.6%
Italian	42,690	1360.8	149.4%	1459	21.5	170.0%
Korean	42,602	1543.8	169.5%	1456	26.2	207.4%
Portuguese	42,696	1264.2	138.8%	1459	19.8	156.5%
Russian	42,671	1503.1	165.1%	1456	24.4	193.1%
Spanish	42,672	1250.5	137.3%	1459	20.7	163.8%

Table 14: Statistics for domain: Aerospace (Part 2 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	57,393	1520.6	165.7%	1339	26.0	193.7%
Chinese	63,398	933.5	-	1492	12.8	—
German	57,655	1394.7	152.0%	1340	22.9	170.7%
English	57,671	917.4	-	1340	13.4	—
French	57,661	1387.6	151.2%	1340	24.2	180.7%
Italian	57,658	1421.0	154.9%	1340	23.7	176.7%
Korean	57,545	1516.9	165.3%	1335	27.0	201.5%
Portuguese	57,649	1319.8	143.9%	1340	21.7	161.8%
Russian	57,643	1577.4	171.9%	1340	26.5	197.6%
Spanish	57,660	1312.5	143.1%	1340	22.7	169.4%

Table 15: Statistics for domain: Agriculture Forestry Animal Husbandry Fishery (Part 3 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	57,327	1550.9	166.8%	1613	26.0	195.8%
Chinese	64,363	930.0	-	1649	11.8	—
German	57,466	1423.2	153.1%	1615	23.1	173.9%
English	57,490	929.6	-	1615	13.3	—
French	57,469	1426.3	153.4%	1615	24.6	185.4%
Italian	57,477	1445.9	155.5%	1615	23.8	179.0%
Korean	57,362	1539.8	165.6%	1615	26.9	202.5%
Portuguese	57,474	1338.9	144.0%	1615	21.9	164.6%
Russian	57,438	1597.3	171.8%	1615	26.6	200.6%
Spanish	57,463	1332.1	143.3%	1608	23.0	172.9%

Table 16: Statistics for domain: Artificial Intelligence Machine Learning (Part 4 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	54,352	1460.8	160.0%	1079	26.2	207.6%
Chinese	57,698	1008.1	-	1419	12.2	—
German	54,918	1358.2	148.7%	1081	23.1	183.2%
English	54,969	913.1	-	1083	12.6	—
French	54,921	1331.2	145.8%	1083	24.2	191.9%
Italian	54,931	1356.8	148.6%	1081	23.6	187.0%
Korean	54,786	1423.8	155.9%	1081	25.8	204.9%
Portuguese	54,927	1255.0	137.4%	1083	21.0	166.8%
Russian	54,904	1435.1	157.2%	1083	25.4	201.6%
Spanish	54,912	1241.2	135.9%	1081	22.0	174.5%

Table 17: Statistics for domain: Automobile (Part 5 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	55,866	1471.7	165.7%	1300	26.0	196.2%
Chinese	62,074	943.6	-	1329	12.3	—
German	55,929	1341.8	151.1%	1299	22.6	170.7%
English	55,948	888.1	-	1303	13.3	—
French	55,933	1312.9	147.8%	1303	24.0	180.6%
Italian	55,931	1367.2	154.0%	1303	23.6	178.0%
Korean	55,846	1452.9	163.6%	1301	27.3	205.7%
Portuguese	55,936	1257.3	141.6%	1303	21.5	162.2%
Russian	55,922	1462.4	164.7%	1303	25.6	192.6%
Spanish	55,932	1251.5	140.9%	1303	22.7	170.9%

Table 18: Statistics for domain: Biomedicine (Part 6 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	61,392	1546.3	167.9%	1406	27.7	200.9%
Chinese	63,133	930.7	-	1254	12.2	—
German	61,546	1405.8	152.7%	1411	23.8	172.9%
English	61,582	920.8	-	1411	13.8	—
French	61,562	1394.0	151.4%	1411	25.2	183.2%
Italian	61,562	1410.9	153.2%	1411	24.0	174.5%
Korean	61,465	1477.3	160.4%	1411	27.4	199.1%
Portuguese	61,562	1318.6	143.2%	1411	22.3	162.2%
Russian	61,545	1571.4	170.7%	1411	27.0	195.8%
Spanish	61,561	1308.2	142.1%	1411	23.3	169.1%

Table 19: Statistics for domain: Computer Communication (Part 7 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	61,435	1512.8	164.0%	1384	25.7	200.5%
Chinese	64,004	931.5	-	1657	11.6	—
German	61,852	1383.2	149.9%	1388	22.7	176.8%
English	61,885	922.6	-	1389	12.8	—
French	61,861	1352.0	146.5%	1387	23.5	183.5%
Italian	61,856	1385.5	150.2%	1388	22.8	177.6%
Korean	61,800	1500.8	162.7%	1387	26.6	207.0%
Portuguese	61,860	1285.5	139.3%	1386	20.9	162.6%
Russian	61,854	1480.0	160.4%	1389	24.6	191.8%
Spanish	61,847	1265.8	137.2%	1389	21.8	169.6%

Table 20: Statistics for domain: Computer Programming Code (Part 8 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	50,902	1515.4	154.0%	615	24.5	198.5%
Chinese	56,802	995.7	-	945	11.7	—
German	51,499	1394.2	141.7%	617	21.6	175.3%
English	51,612	983.9	-	624	12.3	—
French	51,511	1365.0	138.7%	624	22.0	178.6%
Italian	51,544	1383.9	140.7%	618	21.6	175.0%
Korean	51,385	1485.0	150.9%	617	24.5	198.8%
Portuguese	51,540	1284.6	130.6%	624	19.5	158.3%
Russian	51,524	1424.9	144.8%	615	22.8	184.8%
Spanish	51,496	1272.4	129.3%	623	20.5	166.4%

Table 21: Statistics for domain: Current Affairs Government Administration (Part 9 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	60,224	1485.1	159.9%	1270	25.5	185.5%
Chinese	63,941	927.4	-	1280	12.3	—
German	60,374	1442.3	155.3%	1284	24.3	176.5%
English	60,390	928.9	-	1286	13.7	—
French	60,375	1423.4	153.2%	1286	25.3	184.0%
Italian	60,382	1445.6	155.6%	1286	24.5	178.3%
Korean	60,216	1536.2	165.4%	1282	27.8	202.0%
Portuguese	60,375	1329.8	143.2%	1286	22.3	162.6%
Russian	60,356	1604.8	172.8%	1286	27.9	203.3%
Spanish	60,365	1319.8	142.1%	1286	23.2	168.7%

Table 22: Statistics for domain: Electric Power Energy (Part 10 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	59,471	1540.7	167.8%	1283	26.9	199.3%
Chinese	63,487	932.5	-	1535	12.5	—
German	59,557	1433.1	156.0%	1285	24.0	177.8%
English	59,580	918.4	-	1288	13.5	—
French	59,562	1436.6	156.4%	1288	25.6	189.3%
Italian	59,561	1469.3	160.0%	1288	25.0	184.8%
Korean	59,486	1526.2	166.2%	1288	27.5	204.0%
Portuguese	59,568	1346.8	146.7%	1288	22.6	167.4%
Russian	59,551	1612.0	175.5%	1288	27.9	206.6%
Spanish	59,566	1333.6	145.2%	1288	23.5	174.2%

Table 23: Statistics for domain: Film Entertainment (Part 11 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	58,987	1530.5	164.5%	1679	30.6	227.9%
Chinese	63,899	932.2	-	1739	12.6	—
German	59,235	1330.3	143.0%	1687	21.4	159.4%
English	59,250	930.3	-	1687	13.4	—
French	59,236	1320.7	142.0%	1687	22.6	167.9%
Italian	59,236	1330.7	143.0%	1687	21.8	162.0%
Korean	59,079	1579.6	169.8%	1687	27.9	208.0%
Portuguese	59,235	1246.7	134.0%	1687	20.0	148.7%
Russian	59,205	1538.4	165.4%	1684	26.3	196.1%
Spanish	59,213	1241.6	133.5%	1687	21.1	157.3%

Table 24: Statistics for domain: Finance Economics (Part 12 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	60,258	1518.8	163.5%	1315	26.6	193.5%
Chinese	64,184	931.8	-	1118	12.1	—
German	60,373	1443.0	155.4%	1315	24.3	176.4%
English	60,395	928.7	-	1315	13.8	—
French	60,375	1423.6	153.3%	1315	25.5	185.2%
Italian	60,383	1453.1	156.5%	1315	24.7	179.1%
Korean	60,287	1535.0	165.3%	1312	27.5	199.9%
Portuguese	60,382	1338.0	144.1%	1315	22.6	164.0%
Russian	60,358	1569.8	169.0%	1315	27.4	199.1%
Spanish	60,369	1320.5	142.2%	1313	23.3	169.1%

Table 25: Statistics for domain: Fineweb (Part 13 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	69,570	1512.8	163.5%	1717	24.2	192.1%
Chinese	69,998	931.4	-	1218	12.3	—
German	69,918	1361.3	147.1%	1728	20.8	164.9%
English	69,999	925.1	-	1728	12.6	—
French	69,933	1348.3	145.7%	1728	22.1	175.4%
Italian	69,912	1368.2	147.9%	1728	21.5	170.7%
Korean	69,643	1537.2	166.2%	1717	25.8	204.9%
Portuguese	69,939	1271.1	137.4%	1728	19.8	156.8%
Russian	69,838	1506.6	162.9%	1728	24.3	192.5%
Spanish	69,883	1258.1	136.0%	1728	20.7	164.2%

Table 26: Statistics for domain: Fire Safety Food Safety (Part 14 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	29,634	1573.6	166.6%	1294	25.7	196.6%
Chinese	46,710	966.3	-	1195	12.0	—
German	29,687	1455.5	154.1%	1295	23.4	178.7%
English	29,703	944.6	-	1296	13.1	—
French	29,700	1459.0	154.5%	1294	25.3	193.7%
Italian	29,696	1503.4	159.2%	1296	24.9	190.5%
Korean	29,644	1559.7	165.1%	1294	26.7	204.6%
Portuguese	29,696	1385.8	146.7%	1296	22.9	175.0%
Russian	29,684	1630.1	172.6%	1296	27.5	210.8%
Spanish	29,692	1360.0	144.0%	1296	23.3	178.2%

Table 27: Statistics for domain: Game (Part 15 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	44,213	1549.6	163.5%	1276	24.7	197.1%
Chinese	44,119	1100.7	-	1499	12.1	—
German	44,341	1386.4	146.3%	1281	20.4	162.8%
English	44,373	947.7	-	1281	12.5	—
French	44,353	1361.2	143.6%	1281	21.5	171.4%
Italian	44,358	1397.4	147.4%	1281	20.8	166.0%
Korean	44,272	1591.5	167.9%	1278	26.3	210.1%
Portuguese	44,353	1292.4	136.4%	1281	19.3	153.6%
Russian	44,299	1490.8	157.3%	1275	23.0	183.6%
Spanish	44,316	1277.2	134.8%	1281	20.1	160.8%

Table 28: Statistics for domain: Law Judiciary (Part 16 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	59,765	1514.9	163.7%	1287	26.6	195.7%
Chinese	64,276	931.6	-	1354	12.1	—
German	59,954	1453.8	157.1%	1283	24.6	180.4%
English	59,987	925.5	-	1293	13.6	—
French	59,962	1410.8	152.4%	1286	25.6	188.1%
Italian	59,965	1450.2	156.7%	1293	24.9	183.1%
Korean	59,773	1504.3	162.5%	1293	27.8	204.0%
Portuguese	59,959	1321.4	142.8%	1286	22.2	163.2%
Russian	59,950	1592.2	172.0%	1286	28.4	208.8%
Spanish	59,953	1312.7	141.8%	1286	23.2	170.4%

Table 29: Statistics for domain: Literature Emotion (Part 17 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	59,078	1499.8	160.2%	1579	24.3	180.3%
Chinese	64,410	935.4	-	1455	12.4	—
German	59,407	1352.2	144.5%	1586	20.8	154.4%
English	59,439	936.0	-	1586	13.5	—
French	59,415	1343.0	143.5%	1586	22.4	165.7%
Italian	59,410	1348.0	144.0%	1586	21.3	157.7%
Korean	59,189	1557.0	166.3%	1586	26.9	199.1%
Portuguese	59,408	1265.4	135.2%	1586	19.5	144.8%
Russian	59,380	1534.6	164.0%	1581	25.6	189.6%
Spanish	59,402	1257.8	134.4%	1586	20.7	153.5%

Table 30: Statistics for domain: Mathematics Statistics (Part 18 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	62,943	1398.9	149.4%	1032	26.1	186.5%
Chinese	50,306	1014.6	-	977	12.8	—
German	63,178	1307.3	139.6%	1041	24.0	171.6%
English	63,420	936.4	-	1041	14.0	—
French	63,214	1278.4	136.5%	1039	24.6	176.2%
Italian	63,196	1290.9	137.9%	1041	24.0	172.0%
Korean	63,049	1354.1	144.6%	1036	26.5	189.3%
Portuguese	63,201	1221.4	130.4%	1041	22.1	158.3%
Russian	63,188	1380.2	147.4%	1035	26.8	191.9%
Spanish	63,169	1216.4	129.9%	1041	23.1	165.4%

Table 31: Statistics for domain: Medicine Health Psychology Traditional Chinese Medicine (Part 19 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	63,143	1554.0	167.8%	1385	27.5	204.1%
Chinese	64,519	929.2	-	1469	11.6	—
German	63,285	1426.6	154.0%	1385	24.4	180.7%
English	63,314	926.3	-	1385	13.5	—
French	63,297	1420.3	153.3%	1385	25.6	189.9%
Italian	63,292	1433.9	154.8%	1385	24.7	183.2%
Korean	63,197	1505.5	162.5%	1378	27.4	203.7%
Portuguese	63,297	1323.2	142.8%	1383	22.5	166.9%
Russian	63,271	1598.4	172.5%	1385	27.8	206.5%
Spanish	63,288	1320.7	142.6%	1385	23.7	175.7%

Table 32: Statistics for domain: Mining (Part 20 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	47,403	1435.2	165.1%	1319	26.3	195.2%
Chinese	58,159	975.6	-	1474	12.6	—
German	47,500	1329.7	152.9%	1319	23.7	176.1%
English	47,521	869.5	-	1319	13.5	—
French	47,500	1325.8	152.5%	1319	24.8	183.9%
Italian	47,503	1351.8	155.5%	1319	24.0	178.4%
Korean	47,438	1448.9	166.6%	1319	33.9	251.5%
Portuguese	47,510	1254.8	144.3%	1319	22.2	164.6%
Russian	47,497	1495.8	172.0%	1319	27.5	204.3%
Spanish	47,497	1240.2	142.6%	1319	23.0	170.3%

Table 33: Statistics for domain: News Media (Part 21 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	53,784	1497.8	162.1%	1264	25.6	190.8%
Chinese	55,021	1024.2	-	1453	12.1	—
German	53,905	1376.1	148.9%	1270	22.6	168.3%
English	53,931	924.1	-	1271	13.4	—
French	53,918	1376.2	148.9%	1271	24.0	178.9%
Italian	53,922	1391.0	150.5%	1271	23.1	172.0%
Korean	53,785	1549.1	167.6%	1266	28.2	210.5%
Portuguese	53,918	1292.1	139.8%	1271	21.2	157.8%
Russian	53,897	1561.0	168.9%	1271	27.0	201.5%
Spanish	53,915	1278.5	138.3%	1271	22.0	163.8%

Table 34: Statistics for domain: Other Information Services Information Security (Part 22 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	31,329	1571.8	165.4%	839	25.2	201.3%
Chinese	38,947	989.5	-	1044	11.8	—
German	31,360	1494.3	157.2%	839	22.8	182.5%
English	31,375	950.5	-	840	12.5	—
French	31,369	1478.1	155.5%	839	24.2	193.1%
Italian	31,366	1515.0	159.4%	839	23.6	188.7%
Korean	31,341	1566.9	164.8%	833	25.9	207.2%
Portuguese	31,367	1376.3	144.8%	839	21.1	168.4%
Russian	31,365	1635.4	172.1%	839	26.1	208.7%
Spanish	31,368	1359.9	143.1%	839	21.9	174.8%

Table 35: Statistics for domain: Other Manufacturing (Part 23 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	59,176	1472.4	163.3%	1609	25.8	196.9%
Chinese	63,160	928.7	-	1496	12.1	—
German	59,231	1370.3	151.9%	1615	22.9	174.5%
English	59,269	901.9	-	1615	13.1	—
French	59,248	1338.9	148.5%	1615	24.1	183.5%
Italian	59,250	1383.4	153.4%	1615	23.7	180.5%
Korean	59,170	1433.7	159.0%	1615	26.4	201.4%
Portuguese	59,253	1282.2	142.2%	1615	21.7	165.3%
Russian	59,235	1478.8	164.0%	1615	25.6	195.3%
Spanish	59,252	1268.2	140.6%	1615	22.7	172.7%

Table 36: Statistics for domain: Petrochemical (Part 24 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	57,032	1499.3	164.9%	1245	27.0	196.8%
Chinese	63,395	930.0	-	1605	12.7	—
German	57,116	1389.7	152.8%	1245	24.1	175.9%
English	57,150	909.4	-	1245	13.7	—
French	57,131	1393.0	153.2%	1245	25.7	187.0%
Italian	57,133	1421.4	156.3%	1245	24.9	181.3%
Korean	57,034	1487.5	163.6%	1240	27.7	201.9%
Portuguese	57,127	1317.5	144.9%	1245	23.3	169.5%
Russian	57,114	1559.4	171.5%	1243	27.8	202.9%
Spanish	57,116	1302.2	143.2%	1245	23.9	174.2%

Table 37: Statistics for domain: Real Estate Construction (Part 25 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	55,904	1529.9	166.1%	1618	25.2	191.3%
Chinese	63,618	931.3	-	1698	12.0	—
German	56,015	1403.1	152.3%	1618	22.7	172.1%
English	56,034	921.1	-	1618	13.2	—
French	56,024	1385.8	150.4%	1618	23.7	180.0%
Italian	56,018	1429.6	155.2%	1618	23.1	175.7%
Korean	55,928	1544.4	167.7%	1618	26.9	204.1%
Portuguese	56,020	1314.9	142.8%	1618	21.3	161.3%
Russian	55,999	1572.9	170.8%	1617	26.2	198.8%
Spanish	56,015	1307.3	141.9%	1618	22.2	168.7%

Table 38: Statistics for domain: Sports (Part 26 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	55,871	1499.9	165.0%	1372	32.5	238.5%
Chinese	62,963	931.1	-	1429	12.3	—
German	56,023	1314.6	144.6%	1374	22.1	162.1%
English	56,048	908.8	-	1374	13.6	—
French	56,035	1323.3	145.6%	1374	23.8	174.6%
Italian	56,030	1336.0	147.0%	1374	22.9	167.9%
Korean	55,898	1542.0	169.7%	1368	34.4	252.4%
Portuguese	56,029	1246.1	137.1%	1374	21.2	155.8%
Russian	55,991	1519.3	167.2%	1374	33.4	244.9%
Spanish	56,024	1243.1	136.8%	1374	22.4	164.2%

Table 39: Statistics for domain: Subject Education Education (Part 27 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	61,343	1531.2	164.5%	1669	25.5	190.1%
Chinese	64,739	932.0	-	1751	11.4	—
German	61,544	1410.7	151.5%	1672	22.7	169.2%
English	61,577	931.0	-	1672	13.4	—
French	61,554	1408.9	151.3%	1672	24.3	180.9%
Italian	61,549	1432.4	153.9%	1672	23.2	172.7%
Korean	61,466	1549.0	166.4%	1672	27.1	202.3%
Portuguese	61,561	1318.0	141.6%	1672	21.1	157.2%
Russian	61,525	1590.1	170.8%	1672	26.4	197.0%
Spanish	61,551	1308.9	140.6%	1672	22.1	164.4%

Table 40: Statistics for domain: Technology Scientific Research (Part 28 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	61,031	1510.4	162.8%	1398	26.8	196.6%
Chinese	63,950	930.5	-	1594	12.0	—
German	61,220	1396.1	150.5%	1401	23.7	173.7%
English	61,254	927.7	-	1401	13.6	—
French	61,229	1381.2	148.9%	1401	25.0	183.3%
Italian	61,233	1405.4	151.5%	1401	24.1	176.9%
Korean	61,141	1490.1	160.6%	1401	27.4	201.0%
Portuguese	61,220	1311.7	141.4%	1401	22.2	163.0%
Russian	61,201	1550.6	167.1%	1399	27.2	199.7%
Spanish	61,218	1299.2	140.0%	1401	23.3	170.6%

Table 41: Statistics for domain: Tourism Geography (Part 29 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	53,946	1499.7	163.8%	1504	23.6	182.2%
Chinese	62,844	935.1	-	1456	11.7	—
German	54,127	1338.0	146.2%	1504	20.5	157.9%
English	54,144	915.4	-	1504	13.0	—
French	54,127	1346.8	147.1%	1504	22.0	169.7%
Italian	54,124	1363.1	148.9%	1504	21.1	163.2%
Korean	53,981	1524.5	166.5%	1487	25.2	194.7%
Portuguese	54,124	1279.5	139.8%	1504	19.7	152.2%
Russian	54,037	1538.5	168.1%	1503	24.5	188.9%
Spanish	54,099	1277.9	139.6%	1504	20.8	160.2%

Table 42: Statistics for domain: Transportation (Part 30 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	53,730	1516.6	166.7%	1418	26.1	195.6%
Chinese	63,115	931.9	-	1370	12.2	—
German	53,812	1401.2	154.0%	1418	23.4	174.8%
English	53,829	909.9	-	1418	13.4	—
French	53,820	1383.5	152.1%	1418	24.4	182.9%
Italian	53,821	1435.5	157.8%	1418	24.4	182.4%
Korean	53,745	1513.9	166.4%	1415	27.3	204.5%
Portuguese	53,821	1317.5	144.8%	1418	22.0	164.8%
Russian	53,801	1580.0	173.6%	1418	27.3	204.2%
Spanish	53,818	1311.4	144.1%	1418	23.1	172.7%

Table 43: Statistics for domain: Water Resources Ocean (Part 31 of 31).

Language	#Corpus	Avg. Tokens per Document	Ratio (Rel. English)	#Queries	Avg. Tokens per Query	Ratio (Rel. English)
Arabic	53,749	1514.7	166.9%	1328	26.3	195.4%
Chinese	63,027	929.0	-	1489	12.2	—
German	53,839	1403.8	154.7%	1328	23.8	176.6%
English	53,854	907.5	-	1328	13.5	—
French	53,838	1415.5	156.0%	1328	25.4	188.7%
Italian	53,842	1439.7	158.6%	1328	24.8	184.4%
Korean	53,746	1487.7	163.9%	1325	26.7	198.5%
Portuguese	53,841	1325.7	146.1%	1328	22.5	167.0%
Russian	53,813	1579.1	174.0%	1328	27.4	203.2%
Spanish	53,843	1323.7	145.9%	1328	23.5	174.7%